# QUANTITATIVE METHODS FOR DATA SCIENCE

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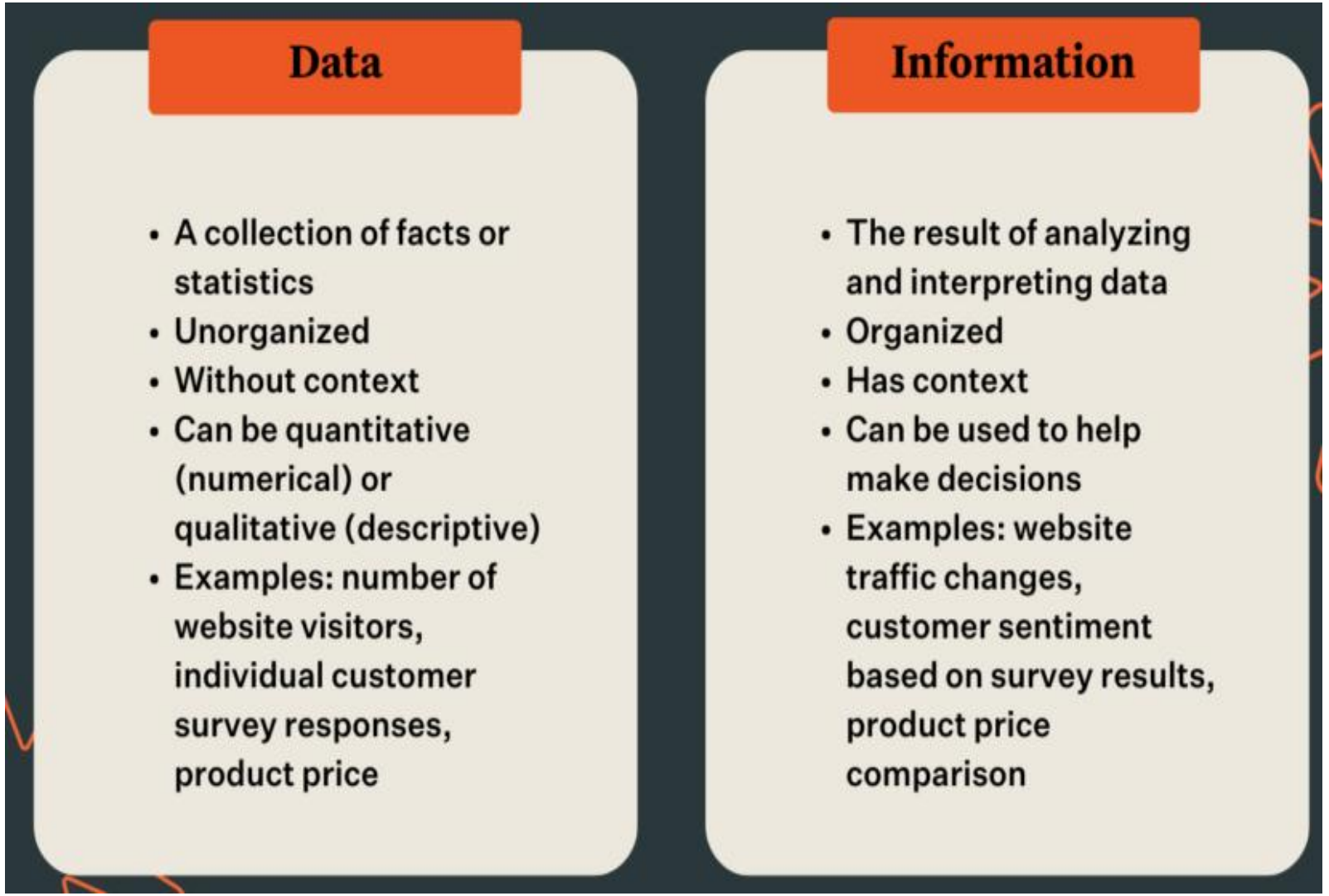
# Quantitative methods

- Problem-solving using data” instead of just opinions

# What is Statistics?

- Statistics converts data into information.

# Data vs Information



# What is Statistics?

- Statistics is a group of methods used to collect, analyse, present, and interpret data and to make decisions.



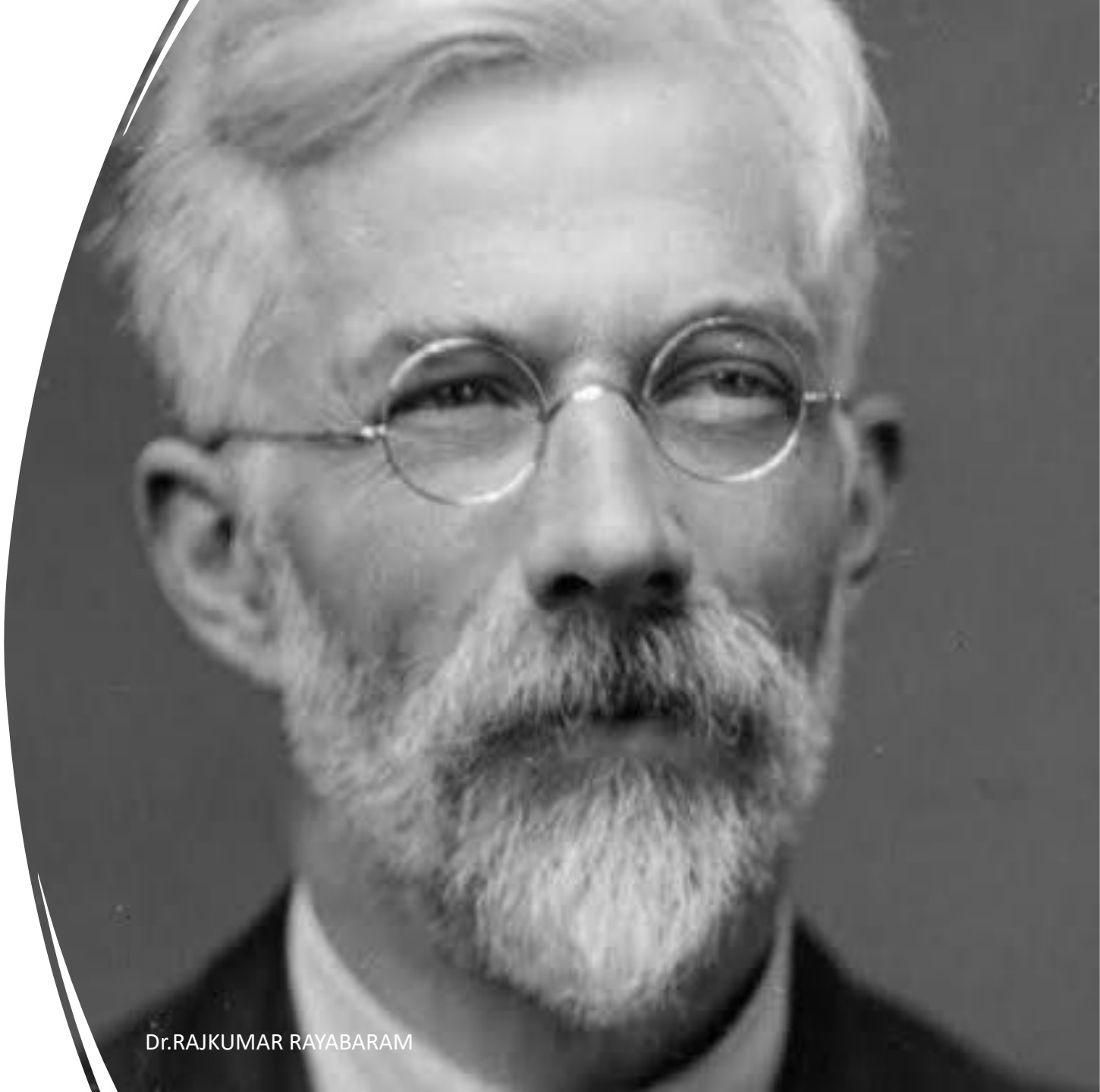

# What is statistics?

HISTORY

- The word statistics was first used by a German scholar **Gottfried Achenwall** in the middle of the 18$^{th}$ century as the science of statecraft concerning the collection and use of data by the state.

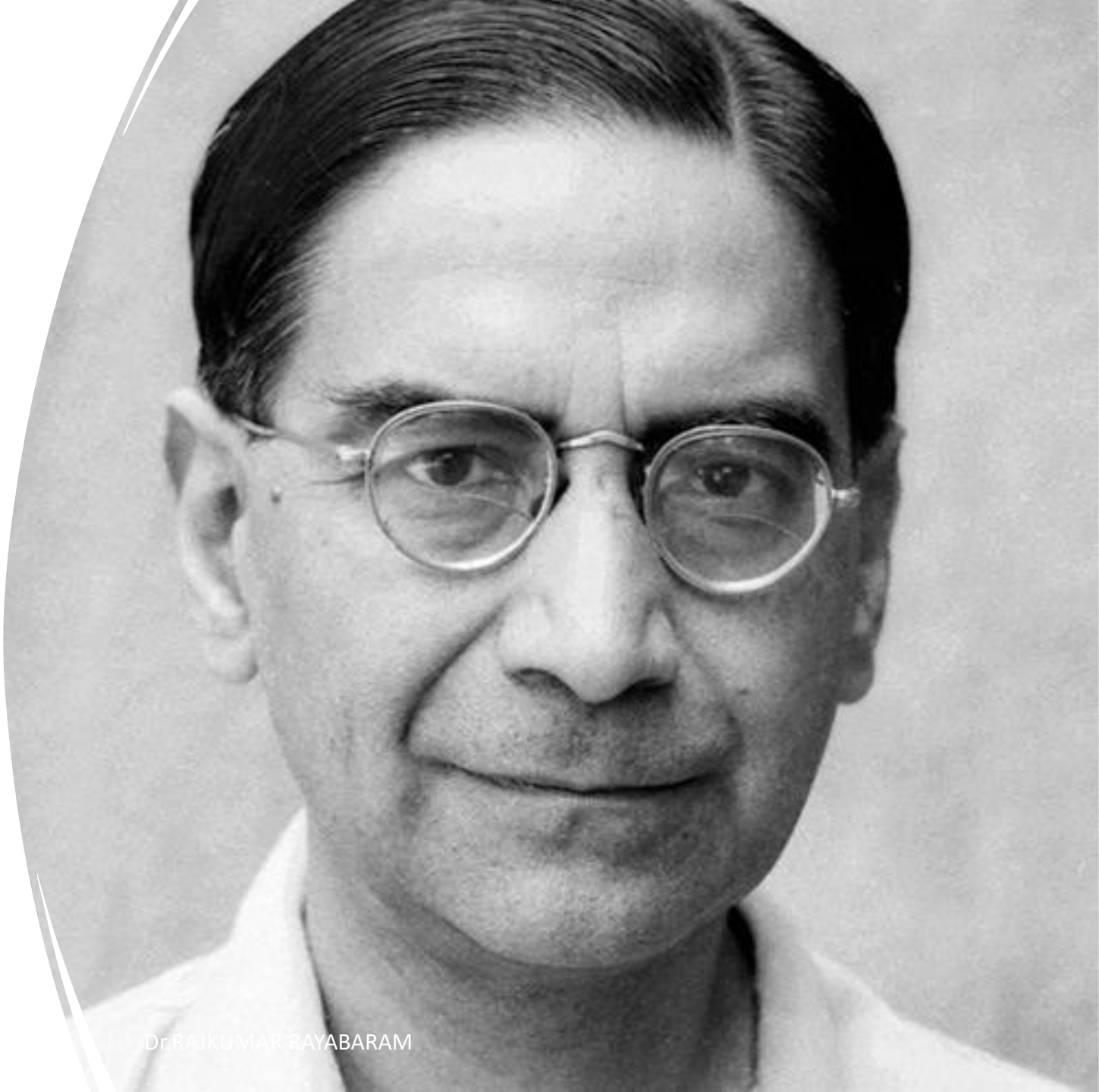
# Father of statistics

Sir Ronald Aylmer Fisher



# Father of Indian Statistics

Prof. Prasanta Chandra Mahalanobis



# The word statistics comes from the

- Latin word “Status” or
- Italian word “Statistia” or
- German word “Statistik” or
- the French word “Statistique”;
- meaning a political state, and originally meant information useful to the state, such as information about sizes of the population (human, animal, products, etc.) and armed forces.



# Types of Statistics

Broadly speaking, statistics can be divided into two areas:

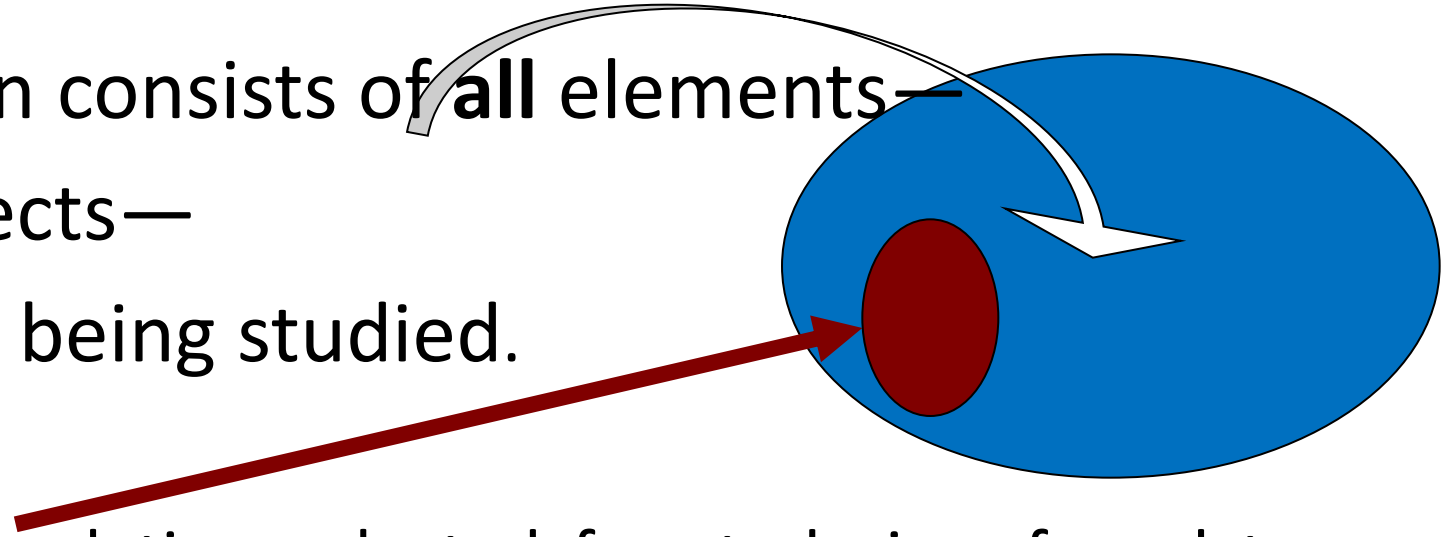
- Descriptive statistics and
- Inferential statistics.
- **Descriptive Statistics** Descriptive statistics consists of methods for organizing, displaying, and describing data by using tables, graphs, and summary measures.
- Descriptive statistics are used in everyday life in areas such as government, healthcare, business, and sport.

- **Inferential statistics** consists of methods that use sample results to help make decisions or predictions about a population.

- Population: A population consists of **all** elements—

individuals, items, or objects—

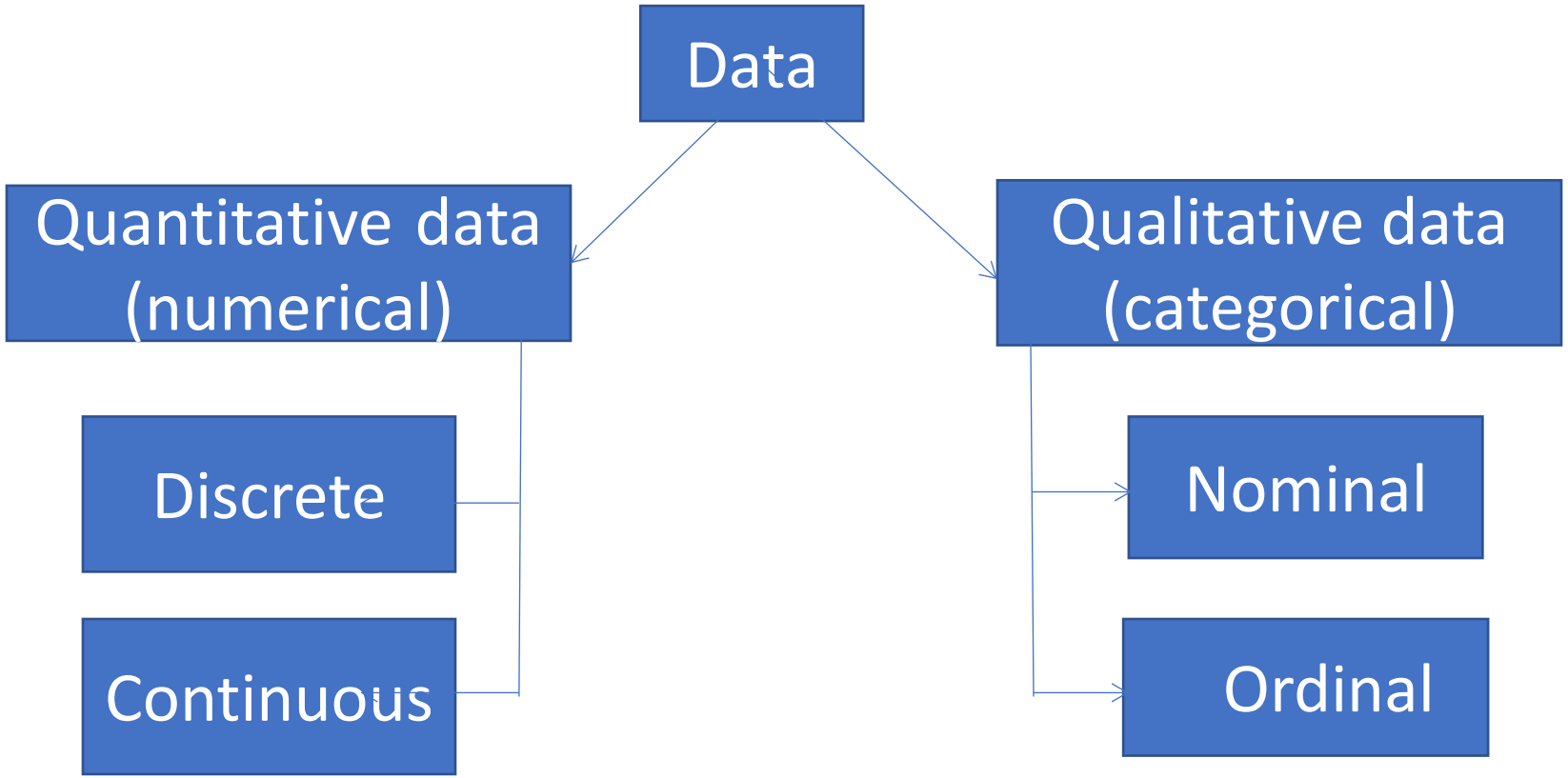
whose characteristics are being studied.



Sample: A portion of the population selected for study is referred to as a sample

# Types of Data

- Data: data are collection of any number of related observations.



- **Discrete**: counted

Discrete Data can only take certain values.

Example: the number of students in a class

- **Continuous**: measured

Continuous Data can take any value (within a range)

Example: A person's height

- **Nominal:** categorised

Without order

Example: Gender (Male, Female, Other.)

- **Ordinal:** categorised

With order

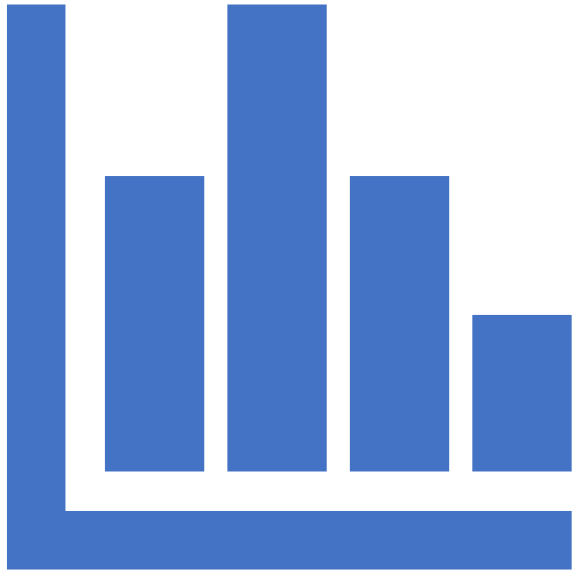
Example: Socioeconomic status (low income, middle income, high income)

# Types of Data in Business Statistics

- Quantitative Data (Numerical):
- Discrete: Countable (e.g., Number of ATMs in a city).
- Continuous: Measurable (e.g., Interest rates, stock prices).

- Qualitative Data (Non-numerical):
- Nominal: Categories without order (e.g., Credit Card Type – Gold, Platinum).
- Ordinal: Ordered categories (e.g., Customer Satisfaction – High, Medium, Low).

# Descriptive statistics



- Descriptive Statistics is a measure that **describes the data.**
- Descriptive statistics consists of methods for organizing, displaying, and describing data by using tables, graphs, and summary measures.
- Descriptive statistics are used in everyday life in areas such as government, healthcare, business, and sport.

# Descriptive Statistics

Types of descriptive statistics:

- Organize Data
  - Tables
    - Frequency Distributions
    - Relative Frequency Distributions
- Displaying Data
  - Graphs
    - Bar Chart or Histogram
    - Scatter Plot or Line
    - Box Plot
- Summarize Data
  - Central Tendency
  - Variation

Below Data collected from the 35 Students Marks in  Maths Subject

70 66 60 55 61 63 72

68 60 60 63 60 75 68

59 71 53 75 64 64 52

64 64 68 64 66 67 63

64 70 69 68 63 59 57

When data are collected in original form, they are called raw data. Raw data is not very meaningful to an audience.

- An array is the arrangement of the values in ascending or descending order.

52 53 55 57 59 59 60

60 60 60 61 63 63 63

63 64 64 64 64 64 64

66 66 67 68 68 68 68

69 70 70 71 72 75 75

# Frequency Distribution

**Organize Data:**

- Organize Data using frequency distributions(Tables)

- A frequency distribution is the organizing the raw data in table form, using classes (groups) and frequencies
  - Class (group) is a quantitative or qualitative category
  - Frequency, *f*, of a class is the number of data values contained in a specific class

52 53 55 57 59 59 60

60 60 60 61 63 63 63

63 64 64 64 64 64 64

66 66 67 68 68 68 68

69 70 70 71 72 75 75

# Discrete frequency distribution

- When a frequency distribution table lists all of the individual categories (X values) it is called a **ungrouped frequency distribution**.
- It can be used when the range of values in the data set is not large.

| Class(X) | Frequency |
|---|---|
| 52 | 1 |
| 53 | 1 |
| 55 | 1 |
| 57 | 1 |
| 59 | 2 |
| 60 | 4 |
| 61 | 1 |
| 63 | 4 |
| 64 | 6 |
| 66 | 2 |
| 67 | 1 |
| 68 | 4 |
| 69 | 1 |
| 70 | 2 |
| 71 | 1 |
| 72 | 1 |
| 75 | 2 |
| **Total** | **35** |
| | |

# Continuous Frequency Distribution

- Sometimes, however, a set of observations covers a wide range of values. In these situations, a list of all the X values would be quite long - too long to be a “simple” presentation of the data. To remedy this situation, a **grouped frequency distribution** table is used.
- **Grouped frequency distributions:** The data must be grouped into classes that are more than one unit in width. In a grouped table, the X column lists groups of observations, called **class intervals**, rather than individual values.

| X | Frequency |
|---|---|
| 52-55 | 3 |
| 56-59 | 3 |
| 60-63 | 9 |
| 64-67 | 9 |
| 68-71 | 8 |
| 72-75 | 3 |
| Total | 35 |
| | |

# Relative frequency distribution

- A relative frequency distribution presents frequencies in terms of percentages.
- Relative frequency of a class

$$= \frac{\text{Frequency of that class}}{\text{Sum of all frequencies}}$$

- Percentage=(Relative frequency )*100

| Class | | Frequency | Percent |
|---|---|---|---|
| | 52-55 | 3 | 8.6 |
| | 56-59 | 3 | 8.6 |
| | 60-63 | 9 | 25.7 |
| | 64-67 | 9 | 25.7 |
| | 68-71 | 8 | 22.9 |
| | 72-75 | 2 | 5.7 |
| | 76-79 | 1 | 2.9 |
| | Total | 35 | 100.0 |

# Organizing Qualitative Data

- **Frequency Distribution for Qualitative Data** A frequency distribution for qualitative data lists all categories and the number of elements that belong to each of the categories.

| Course | No.of Students |
|---|---|
| BBA | 10 |
| BBA BA | 6 |
| BBA FM | 7 |
| BCOM | 8 |
| BBA MA | 4 |
| SUM | 35 |